MAT216: Linear Algebra and Fourier Transformation

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LECTURE-02

Table of Contents

1 Gaussian Elimination

Different taste!

$$\begin{cases} x-y &= 1 \\ 2x+y &= 6 \end{cases} \qquad \begin{cases} x+y &= 4 \\ 3x+3y &= 6 \end{cases} \qquad \begin{cases} 4x-2y &= 1 \\ 16x-8y &= 4 \end{cases}$$

System 1: "We'll end up in mutual understanding" $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

System 2: "We'll never intersect!" $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

System 3: "Wait, Are we different?" $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

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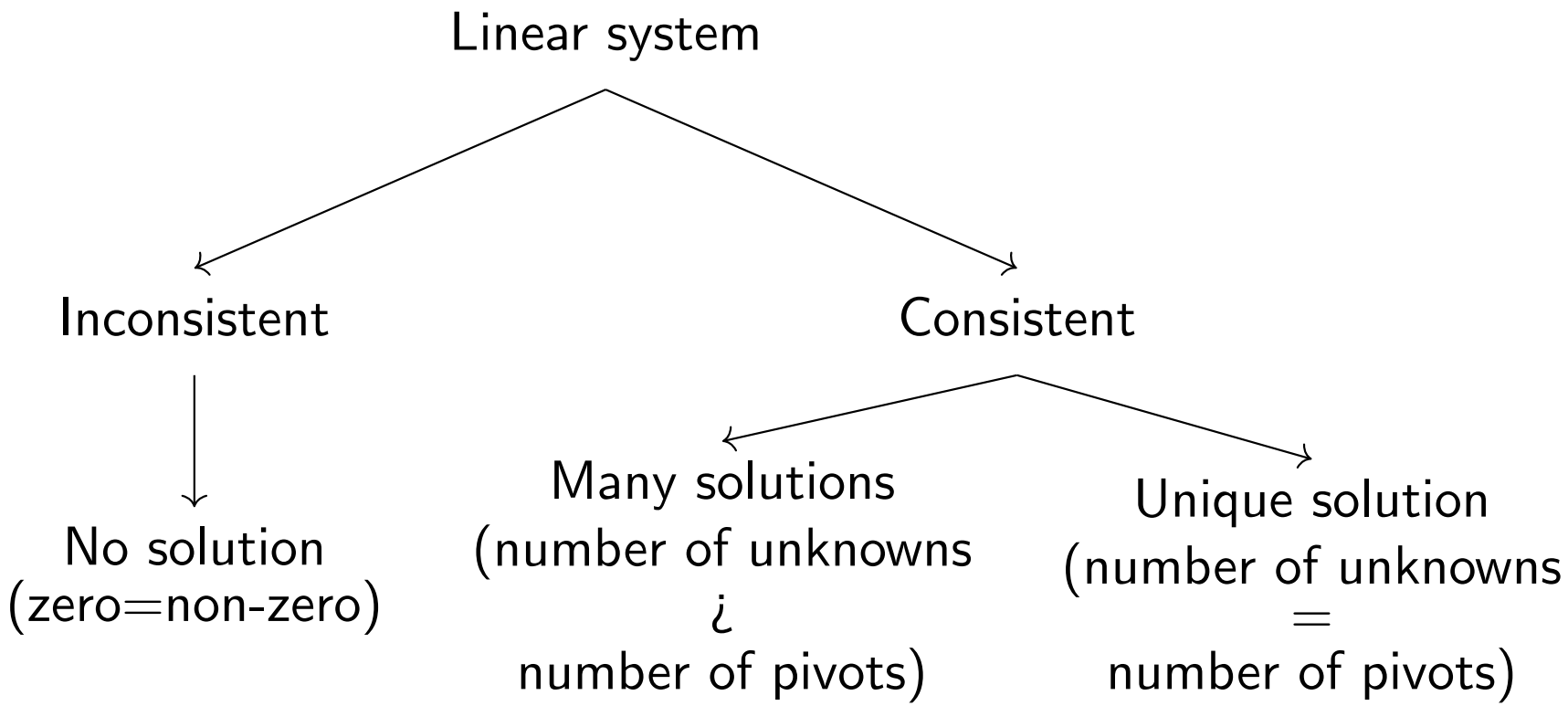
Observation: The First system gives a unique solution, the second one gives no solution ($0 = -6$) and the third one gives infinitely many solutions ($0 = 0$).

Takeaway:

(1) If your system contains $0 = \star$ then the system is inconsistent.

(2) And if your system contains $0 = 0$ then it is consistent (with infinitely many solutions).

Tree


Linear system
Inconsistent
Consistent
No solution
(zero=non-zero)
Many solutions
(number of unknowns
¿
number of pivots)
Unique solution
(number of unknowns
=
number of pivots)

Some more systems!

Identify the nature of the solutions of the following systems:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \qquad \left(\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{ccc|c} 1 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Reverse engineering

Solve the system from the nicer form:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 8 \end{array}\right) \qquad \left(\begin{array}{ccccc|c} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 7 \\ 0 & 0 & 1 & 8 \end{array}\right)$$

Operation tools!

We call a column as a Pivot column if it has only one non-zero entry and rest of them are zeros. As for the 3×3 matrix, we can have the following scenarios:

$$\begin{pmatrix} \star \\ 0 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ \star \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 0 \\ \star \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} \star & \star & \star & \star \\ \star & \star & \star & \star \\ \star & \star & \star & \star \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ \star & \star & \star & \star \\ \star & \star & \star & \star \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ 0 & \star & \star & \star \\ 0 & \star & \star & \star \end{array}\right)$$

$$\rightarrow \left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ 0 & 1 & \star & \star \\ 0 & \star & \star & \star \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ 0 & 1 & \star & \star \\ 0 & 0 & \star & \star \end{array}\right) \rightarrow \underbrace{\left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ 0 & 1 & \star & \star \\ 0 & 0 & 1 & \star \end{array}\right)}_{\text{Row Echelon form}}$$

You can use back substitution or row operations to get the solution.

continued...

$$\left(\begin{array}{ccc|c} 1 & \star & \star & \star \\ 0 & 1 & \star & \star \\ 0 & 0 & 1 & \star \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & \star & \star \\ 0 & 1 & \star & \star \\ 0 & 0 & 1 & \star \end{array}\right) \to \underbrace{\left(\begin{array}{ccc|c} 1 & 0 & 0 & \star \\ 0 & 1 & 0 & \star \\ 0 & 0 & 1 & \star \end{array}\right)}_{\text{Reduced Row Echelon form}}$$

Now, Apply your tools to the following system:

$$\begin{cases} x+y+2z & =9 \\ 2x+4y-3z & =1 \\ 3x+6y-5z & =0 \end{cases} \Longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right)$$

continued...

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continued...

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continued...

Gaussian Elimination: https://www.youtube.com/watch?v=bnC848ie16Q

